

Integration — Lesson 31 Test: Integration by Parts II

10 questions · show your work in the box · every indefinite integral needs "+ C" · exact answers (leave e , π , $\ln 2$ as symbols)

Name: _____ Date: _____ Score: _____ / 10

1. Find $\int x e^{2x} dx$. (One pass of parts; remember the antiderivative of e^{2x} is $e^{2x}/2$.) 1 pt

2. Find $\int 3x \cos x dx$. 1 pt

3. Decide: u-substitution or integration by parts? Then solve it: $\int x \sin(x^2) dx$. 1 pt

4. Find $\int (x^2 + 1) e^x dx$. (Hint: split it into two integrals; one of them needs parts twice.) 1 pt

5. Find $\int \ln(2x) dx$ using parts with $dv = dx$.

1 pt

6. Find $\int x^3 \ln x dx$. (LIATE: which factor must be u ?)

1 pt

7. Compute $\int_1^{e^2} \ln x dx$ exactly. (Recall $\ln(e^2) = 2$ and $\ln 1 = 0$.)

1 pt

8. Compute $\int_0^{\pi/2} x \cos x dx$ exactly.

1 pt

9. Compute $\int_0^1 x e^x dx$ exactly. (The answer is surprisingly clean!)

1 pt

10. Challenge: find $\int x^3 \cos x dx$ using the tabular method. Show your D column, your I column, and the diagonal products with alternating signs.

1 pt
